EXPERIMENT 13: IMPLEMENT YOUR OWN PING PROGRAM

AIM

To implement a simple ping program to check whether a destination host is reachable over a network.

THEORY

Ping checks connectivity by sending an Internet Control Message Protocol (ICMP) Echo Request to a destination and waiting for an Echo Reply. The reply time gives an approximate round-trip time. A timeout may indicate that the host is unreachable or that ICMP traffic is blocked.

PROGRAM (Python)

The program sends an ICMP Echo Request using the system ping command and displays the result. Save as `my_ping.py`.

```
import subprocess
import platform

def ping(host):
    # Windows uses -n; Linux/macOS use -c
    option = "-n" if platform.system().lower() == "windows" else "-c"
    result = subprocess.run(
        ["ping", option, "4", host],
        capture_output=True,
        text=True
    )
    print(result.stdout)
    if result.returncode == 0:
        print(f"{host} is reachable.")
    else:
        print(f"{host} is unreachable or not replying.")

if __name__ == "__main__":
    destination = input("Enter host/IP to ping: ").strip()
    ping(destination)
```

PROCEDURE

1. Save the program as `my_ping.py`.
2. Open a terminal in the directory containing the file.
3. Run `python3 my_ping.py`.
4. Enter a hostname such as `google.com` or a reachable IP address.
5. Observe the Echo Reply messages and packet statistics.

SAMPLE OUTPUT

```
Enter host/IP to ping: 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

127.0.0.1 is reachable.
```

OBSERVATIONS

The program tests host reachability by invoking the operating system's ping utility. Successful replies indicate that ICMP Echo traffic is reaching the destination and returning to the sender.

RESULT

A simple ping program was implemented in Python and used to check destination host reachability.